SESSION 2 — AFTERNOON

Macroeconomics & Global Finance

GDP, cycles, inflation, policy and exchange rates

Learning Objectives

GDP & National Accounts

Define and calculate GDP using expenditure approach. Distinguish nominal vs real GDP.

Business Cycles

Identify cycle phases and their implications for asset allocation and sector rotation.

Inflation & Unemployment

Explain CPI, PPI, GDP deflator, Phillips curve. Apply Fisher equation to nominal/real returns.

Monetary & Fiscal Policy

Describe central bank tools (rates, QE, forward guidance) and government policy levers.

Gross Domestic Product (GDP)

$$GDP = C + I + G + (X - M)$$

C	**Consumption** Household spending on goods & services. Typically 60-70% of GDP in developed economies.	*UK: consumer spending on food, housing, healthcare, recreation*
I	**Investment** Business capital expenditure + residential construction + inventory changes.	*Company building new factory, households buying homes*
G	**Government** Government spending on goods & services (not transfers like benefits).	*NHS spending, defence, infrastructure, public sector wages*
X−M	**Net Exports** Exports minus imports. Negative if country imports more than it exports (trade deficit).	*UK has persistent trade deficit (X < M), so X−M is negative*

Nominal vs Real GDP — Stripping Out Inflation

Nominal GDP is measured in current prices — it rises when output grows AND when prices rise. Real GDP adjusts for inflation.

GDP Deflator = (Nominal GDP / Real GDP) × 100 → Real GDP = Nominal GDP / Deflator × 100

WORKED EXAMPLE

Year 1: Nominal GDP = £2,000bn, GDP Deflator = 100 (base year) → Real GDP = £2,000bn
Year 2: Nominal GDP = £2,200bn, GDP Deflator = 105 → Real GDP = 2,200/105 × 100 = £2,095bn

Nominal growth = +10%. Real growth = +4.75%. Inflation accounted for = 5.0%.

When Nominal > Real Growth

Inflation is positive. Economy growing but purchasing power rising slower than headline suggests. Watch for real returns!

Per-Capita GDP

GDP / Population = living standards proxy. A country's GDP can grow while per-capita GDP falls if population grows faster.

Business Cycles — The Investment Cycle

Economic activity fluctuates in recurring patterns. Asset allocation varies dramatically by cycle phase.

Expansion

GDP rising, employment improving, credit expanding, consumer confidence high. Inflation begins to build.

→ **Equities, high-yield bonds, commodities, cyclical sectors**

Peak

Activity at highest point. Inflation elevated, central banks tightening. Leading indicators start turning.

→ **Inflation-linked bonds, commodities, short-duration assets**

Contraction

GDP falling (recession if ≥2 consecutive quarters), unemployment rising, credit tightening, confidence collapses.

→ **Government bonds, defensive equities, cash**

Trough

Activity bottoms, policy turns accommodative, early green shoots emerge. Recession ends here.

→ **Equities (early movers), high-beta stocks, cyclicals**

Inflation — Measuring & Understanding Price Changes

Inflation is a persistent rise in the general price level. It erodes real returns and affects every asset class.

Fisher Equation: Nominal return ≈ Real return + Inflation rate → Real r ≈ Nominal − Inflation

EXAMPLE — Fisher Equation
A bond yields 5% nominal. Inflation is 3%.
Real return ≈ 5% − 3% = 2%
Exact formula: (1 + nominal) = (1 + real)(1 + inflation) → Real = (1.05/1.03) − 1 = 1.94%

CPI (Consumer Price Index)

Tracks a basket of household goods & services. Most commonly cited inflation measure. Published monthly.

PPI (Producer Price Index)

Tracks factory-gate prices. Leading indicator — PPI rises tend to flow through to CPI with a lag.

Demand-Pull Inflation

Cost-Push Inflation

Monetary Policy — Central Bank Toolkit

Central banks manage price stability and employment through the price and supply of money.

Policy Rate (Base Rate)

Sets overnight borrowing rate. Transmission: influences all other rates via the yield curve.

↑ Rate → ↑ borrowing cost → ↓ spending → ↓ inflation. ↓ Rate → stimulate.

Quantitative Easing (QE)

Central bank buys government/corporate bonds → injects reserves → lowers yields.

Lowers long-term rates. Inflates asset prices (wealth effect). Controversial — distributional effects.

Forward Guidance

Signalling future rate intentions. 'Rates will stay low until unemployment falls below X%.'

Shapes expectations before acting. Can move markets as powerfully as rate changes.

Reserve Requirements

Minimum % of deposits banks must hold. Rarely changed in developed markets.

Lowering requirements → more lending capacity (money multiplier expands).

Exchange Rates — PPP and Interest Rate Parity

The exchange rate links countries' economies. Two key theories explain how rates should behave over time:

Purchasing Power Parity (PPP)

Law of one price applied globally: identical goods should cost the same everywhere when expressed in a common currency.

Forward FX rate / Spot rate ≈ (1 + π_foreign) / (1 + π_domestic)

Countries with higher inflation → currency depreciates.

Big Mac Index (Economist): informal PPP test. If a Big Mac costs £4 in UK and $6 in US, but GBPUSD = 1.40, the pound is fairly valued.

Best for: long-run FX forecasting, cross-country valuation.

Interest Rate Parity (IRP)

Higher interest rate currencies depreciate by the interest rate differential. Prevents risk-free arbitrage.

Forward rate = Spot × (1 + r_d) / (1 + r_f)

Ex: UK rates 5%, US rates 3%. £ should depreciate ~2% vs $ to prevent arbitrage.

Covered IRP: holds in practice due to arbitrage enforcement.

Uncovered IRP: a forecasting theory — often fails short-term (carry trade exists!).

Carry trade: borrow in low-rate currency, invest in high-rate — profitable until it isn't.

PRACTICE PROBLEMS

Try before turning the page.

Q1	Country A: Nominal GDP £2,400bn, GDP Deflator = 120. Calculate Real GDP. If last year's real GDP was £2,250bn, what is real growth?
Q2	A government bond yields 6%. Inflation is 2.5%. Calculate the real return using both the approximation and exact Fisher formula.
Q3	An economy moves from Expansion → Peak. Which THREE asset classes would you overweight / underweight and why?
Q4	Central bank raises rates from 3% to 4%. Walk through the full transmission mechanism to inflation (5 steps).
Q5	UK interest rate = 4%, Eurozone rate = 2%. Current GBPEUR = 1.18. What should the 1-year forward rate be per IRP?

Practice — Answers

A1 Real GDP = 2,400/120 × 100 = £2,000bn. Real growth = (2,000−2,250)/2,250 = −11.1% (recession!)

A2 Approximation: 6% − 2.5% = 3.5%. Exact: (1.06/1.025) − 1 = 3.41%. Difference small but exact is always more precise.

A3 Overweight: inflation-linked bonds (TIPS), commodities, short-duration assets. Underweight: long-duration bonds (most exposed to tightening), rate-sensitive sectors (utilities, REITs).

A4 (1) Higher base rate → (2) ↑ borrowing costs (mortgage, corporate) → (3) ↓ household spending & business investment → (4) ↓ aggregate demand → (5) ↓ inflationary pressure. Works with 12-18 month lag typically.

A5 Forward = 1.18 × (1.04/1.02) = 1.18 × 1.0196 = 1.2031. £ depreciates vs € by ~2% (higher UK rates → forward £ weaker).

KEY TAKEAWAYS

Session 4 — Macroeconomics & Global Finance

- GDP = C+I+G+NX — the economy's scorecard. Real GDP strips out inflation.
- Business cycle phase determines asset allocation — know where you are
- Fisher: Real return = Nominal − Inflation. Always work in real terms.
- Central banks use rates + QE + guidance — transmission takes 12-18 months
- PPP: high inflation → currency depreciates long-run
- IRP: higher rates → forward currency depreciation (prevents arbitrage)